Quiz 5

October 2, 2026 Name: _____

Show your work. **Circle your answer.** **Total: 10 points**

1. Domain and a hole [4 points]

Let

$$R(x) = \frac{x-2}{x^2-4}.$$

(a) Factor the denominator. Give the original domain in interval notation. **[1 point]**

(b) Simplify R while keeping its restrictions. Find the hole as an ordered pair. **[2 points]**

(c) Find all x- and y-intercepts, or state that none exist. **[1 point]**

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2. Asymptotes and limits [6 points]

Let

$$H(x) = \frac{3x + 2}{x + 1}.$$

(a) Find the vertical and horizontal asymptotes. Explain each. **[2 points]**

(b) Find $\lim_{x \to -1^-} H(x)$ and $\lim_{x \to -1^+} H(x)$. **[2 points]**

(c) Find $\lim_{x \to -\infty} H(x)$ and $\lim_{x \to \infty} H(x)$. **[2 points]**